

Punith P

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Summary

Frontend Developer and Software Engineer who has shipped published, real-time 3D web applications and full-stack systems: a Fabric Minecraft mod on CurseForge and Modrinth across three versions, including a server-authoritative security redesign; a WebGL astronomical simulator with real-time procedural skies and geolocation-synced star fields; and a MERN-stack medical records platform with role-based access control and full audit logging. Directs AI-assisted development end-to-end — system architecture, implementation, and debugging — across all projects rather than manual line-by-line coding.

Education

Bachelor of Engineering (B.E.) in Computer Science and Engineering

Sep 2023 – Jun 2027

K S Institute of Technology, Bengaluru

Skills

- **Languages:** Java, JavaScript (ES6+), TypeScript, HTML5, CSS3
- **Frontend:** React, React Three Fiber, Zustand, Tailwind CSS, Responsive Design
- **Backend:** Node.js, Express.js, REST APIs, JWT Authentication, Role-Based Access Control (RBAC), MongoDB, Mongoose, Multer
- **Graphics:** Three.js, WebGL, GLSL, Procedural Generation, Blender
- **Tools & Build:** Git, GitHub, Gradle, Vite, Fabric API

Experience

Lipi — Fabric Minecraft Mod

CurseForge | GitHub

Java | Fabric API | Custom Client-Server Networking

- Redesigned the identity-verification flow to be fully server-authoritative across three releases, closing a client-trust security flaw uncovered during testing
- Built a dedicated in-game chat UI — channel sidebar, localized timestamps, unread tracking, command auto-suggestions — replacing an earlier vanilla-chat-dependent version
- Rolled out server-side moderation: persistent logging, auto-expiring temporary mutes, per-player cooldowns, and GDPR-conscious data handling
- Published on CurseForge and Modrinth under an MIT license, incorporating real external tester feedback to prioritize version support

JeevanLocker — Medical Document Management

GitHub

MERN Stack | TypeScript | JWT | RBAC

- Led a 4-person team as Team Lead for JeevanLocker's IET-sponsored Student Research Project Proposal (SRPP), planning system architecture and directing implementation across the team
- Architected JWT authentication, bcrypt hashing, and role-based access control (Patient/Doctor/Admin) for a full-stack medical-records platform
- Built a document upload and sharing pipeline (Multer) with server-side permission checks — doctors access only explicitly-shared documents — plus audit logging across five event types (login, upload, preview, download, share)

Low-Poly Time Sync Environment

Live Demo | GitHub

Three.js | WebGL | GLSL | SunCalc

- Engineered a GPU-tiered, rejection-sampled star field rendering up to ~7,200 stars from an 18,000-candidate pool, driven by live geolocation and real sun/moon positioning
- Developed five custom GLSL shaders — sky dome, moon-phase, star field, dust, volumetric haze — plus FBM procedural clouds for continuous day-night transitions
- Deployed as a zero-build, native ES-module app on GitHub Pages with adaptive GPU detection and full mobile support

Procedural Low-Poly World Generator

GitHub

React Three Fiber | Three.js | Procedural Generation

- Constructed a seed-based terrain system with three archetypes (alpine, rounded massif, plateau), multi-layer domain warping, and deterministic seeded generation
- Automated instanced vegetation placement via slope, altitude, and treeline sampling, plus neighbor-based ledge detection
- Diagnosed a main-thread freeze in the weather system to a synchronous 16,641-vertex noise computation, and scoped a Web Worker offload fix

Additional Information

- **Achievements:** JeevanLocker was approved for the Institution of Engineers (India) Students Research Project Proposal (SRPP) 2025–26; TCS Tech Bytes statewide engineering IT quiz participant (600+ participants)
- **Certifications:** AI Skills Passport (EY & Microsoft), AWS Cloud Practitioner (Infosys Springboard)
- **Activities:** IEEE ImpactX Hackathon, Generative AI Workshop, IBM SkillsBuild Blockchain & Cybersecurity Training